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1. HyCAT ASV

HYCAT is a man-portable ASV with a broad range of data and survey functionality. Equipped with HYPACK Max, the ASV capabilities include data acquisition, survey planning, general bathymetry, cartography, water quality, discharge measurements, habitat mapping, pipeline surveys, infrastructure inspections, search and rescue and rapid disaster response. Sensors have plug-and-play capabilities, allowing HYCAT to be customized to your specific needs.

HYCAT does not require a ramp or dock launch access and can be easily deployed in remote locations by two individuals. Developed in partnership with Xylem, HYCAT autonomous surface vehicle fits in a standard sized SUV for easy transportation to and from launch sites. The robust ASV has full day survey endurance and navigates remote, shallow, inland and coastal areas easily.

Specifications

Dimensions: Length: 5.9 ft (1.8m) / Beam: 2.83 ft (.86m) / Draft: 0.5ft (.15m)

Weight: 53 kg (115.5 lbs)

Speed Range (survey speed): ASV: 0 – 7.8 kts

Endurance: 8.0 hrs @ 2 kts / 6.0 hrs @ 3 kts / 2.7 hrs @ 4 kts

Operational Sea State Beaufort Wind Scale Identifier 4

Environmental Operating Conditions Air Temperature: -20 to + 45 C / Water Temperature: +4 C to + 32C

Propulsion System: 2x 1KW BLDC Pocket Thrusters

Steering: Differential Thrust

Battery: 1 x 1500 Whr 24VDC nom UN38.3 rated Li-ion Battery. Field swappable.

On-Board Electronics: 6th Gen Intel Core i5 GHz processor with MS Windows 10. 512 GB (Solid State Hard Drive).

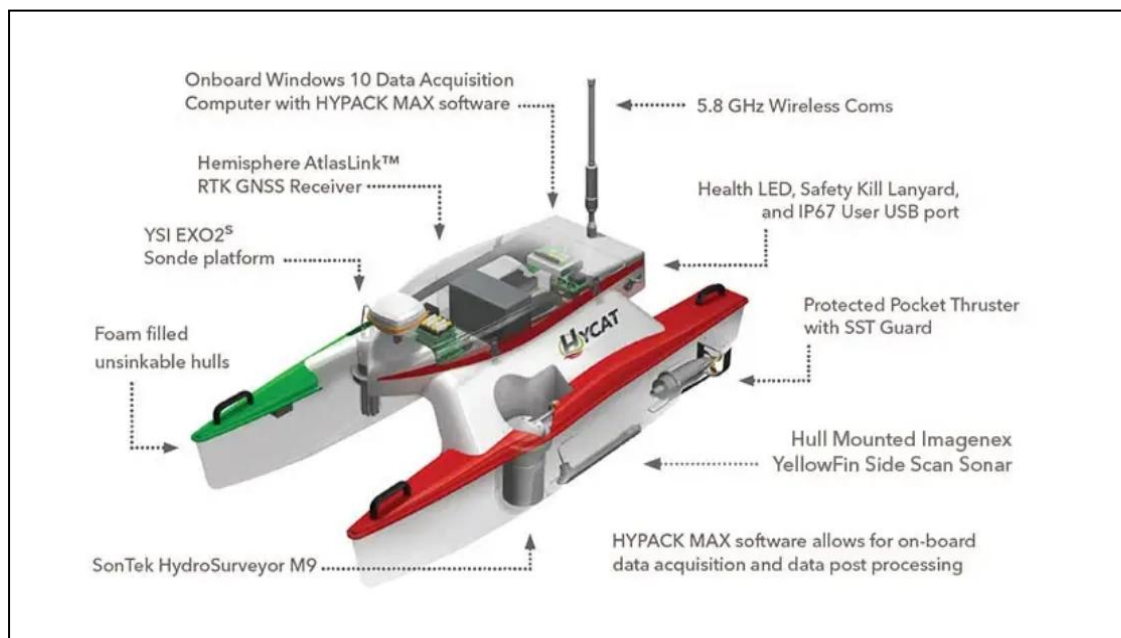
Mission Planning Tools: REALTIME mission planning, monitoring and data acquisition via ASV driver for HYPACK MAX.

Communications: Line of Sight Wi-Fi; 5.8GHz

Handheld Remote Controller: Ruggedized operator control unit with internal DC and external 120/240 VAC external power input. Joystick with softkeys to effortlessly switch between autonomous and RC control modes.

Software: HYPACK MAX, SonTek Hydro Surveyor, YSI EXO KOR

The HYCAT Autonomous Surface Vehicle has the following components.



Important steps for to be noted before operating the boat:

- Before turning on the boat, make sure all the instruments and sensors on the boat are installed and connected, including the radio antenna on boat.
- Check the battery connection and make sure that it is fully charged. Fully charged battery can power the boat for about 10 hours
- Check for the USB data thumb drive and the USB dongle key inserted to the HyCAT boat USB ports.

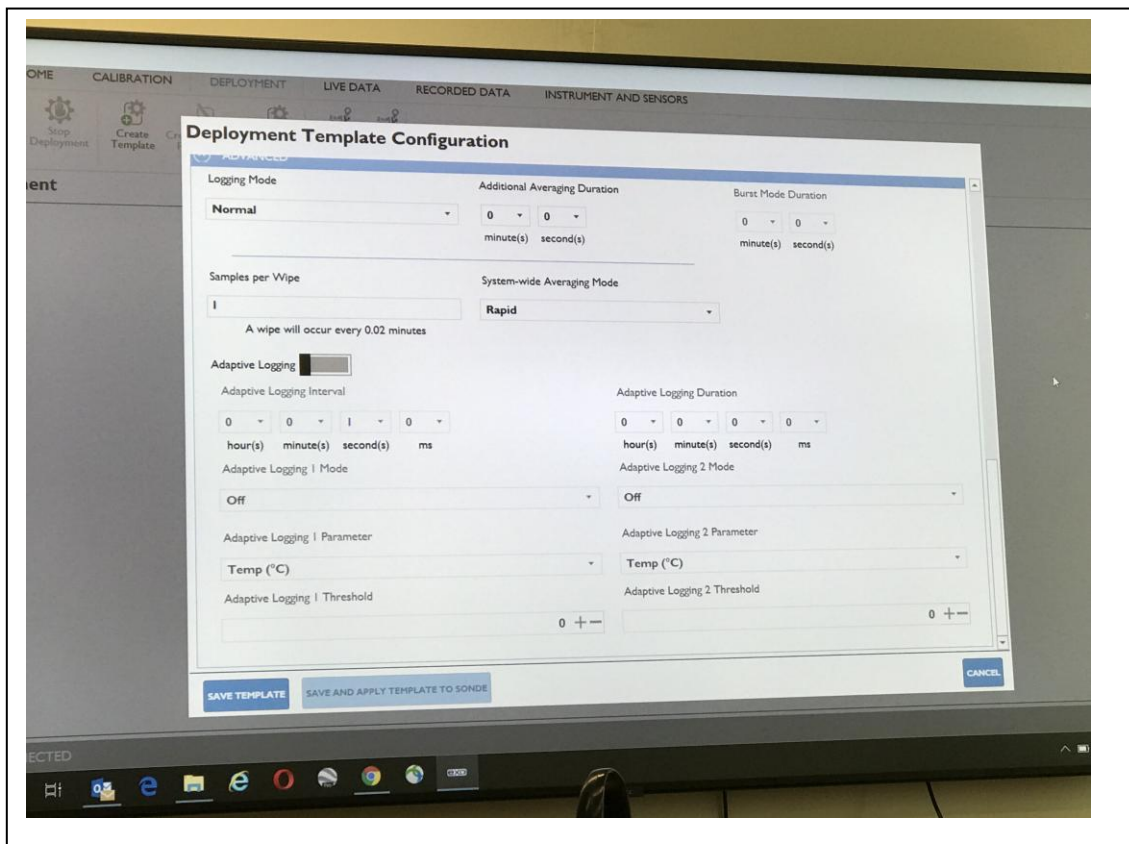
- Check propellers to make sure that both are rotating properly, and no mud, tree branches or other debris are stuck in it.
- The red cap must be closed and attached, before launching the boat into the water. Otherwise, **you will not be able to control and operate the boat!**
- YSI EXO Sonde, SonTek HydroSurveyor M9, and GARMIN GPS are installed and connected properly.
- Connect/start the HyCAT boat by quickly pressing the mode button 3 of the controller twice and joystick button 1 once
- Boat operation speed: 2 knot (1.03m/s)
- Radio signal range: ~1 km

2. YSI EXO Sonde Preparation and Installation on the HyCAT

- Use KOR software computer program for YSI EXO3 to set deployment mode
- KOR is free and can be installed on any computer
- Either use USB plug to connect computer to YSI sonde or use magnetic key to turn on YSI sonde and use blue tooth to connect computer KOR with YSI sonde
- Tutorial videos for EXO and KOR are available at

<https://video.ysi.com/ysi-university-exo>

- Go to Deployment/advanced in KOR



Average: 40 sec

Accelerated: 5-10 sec

Rapid: 2- 3 sec

Select Rapid mode

- Select parameters for dsl output, and in HyPACK the water quality parameters need to be selected in the same order as defined in KOR
- Do calibrations

3. Radio Base Station

Radio station is established to keep radio communication between the controller and HyCAT boat. The effective radio communication range is about 1 km, when no obstacles are present in the line of sight between the radio antenna and the HyCAT boat.

The radio Base Station must be connected with the Black hub box to get the power. Generally, a long pole is used to place the radio antenna high so that the radio signal can reach and cover a wide range.

- If the connection of the Base and HyCAT boat is lost, then we cannot manually control the HyCAT in real time, it will move according to the previously programmed routes by the user.
- Proper planning is required so that we can maintain the boat is constantly connected.
- Make sure all the blue lights of radio antenna are on, which indicate the proper connection with the HyCAT boat.



4. Blackbox Hub

The radio station, the laptop server computer, TV display monitor, and power source are connected through a black box hub.

- Black box has the several ports: Radio, Laptop Ethernet, RTK, User Ethernet, and Power Supply.

- Laptop Ethernet port will be used to connect with the laptop Server Computer,
- Radio port will connect to the Base Radio station antenna.
- The extra User Ethernet can be used to connect one more computer if it is needed.
- The RTK port is available, as our HyCAT boat only uses a GARMIN GPS, rather than expensive RTK GPS.
- Make sure all the cables are firmly connected. Any loose connections may result in operational failure.
- The Black Box hub must be connected to a stable power supply source.



Connection of Radio, Laptop and Power to the Blackbox hub

5. Connections between different components and boat launching

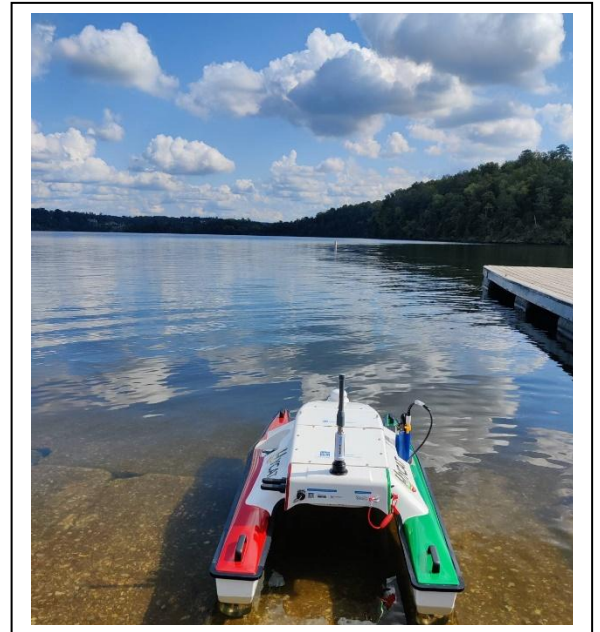
After the Base Radio station, the laptop computer, the blackbox hub, power supply source are connected, turn on the power for every component. It is advisable to turn on the power in the following order:

- Power on the HyCAT Boat first, and it may take about 3 minutes.
- Power on the Radio Receiver and extend to the desired height, and it takes about 30 seconds.
- Power on the Black Box hub.
- Make sure your laptop computer is powered on and connected with the controller via USB drive plugged in.
- The green blue lights on the Radio antenna indicate if the Radio Base station and the HyCAT boat are properly connected.

Before boat launch, please check the following:

- Power supply is needed for laptop computer and TV monitor (major drain for power)

- Remote Controller needs AA batteries. Its connection with laptop computer is through a small USB plug to the laptop computer
- Remember to put the USB Thumb data and USB HyPACK dongle key in the USB ports and tight the cover caps.
- Check boat propellers to make sure they can freely spin and no debris is in it
- The red cap must be closed and attached before we start the boat engine and launch it into the water. Make sure it is attached before we start the boat.
- Establish emergency return point for boat and calibrate the compass on the boat.
- Remember to set the Emergency Return Point before performing any mission.
- Remember to calibrate the compass on the boat before the mission



6. Laptop Server Computer and software

The laptop computer that is used by the operator is referred as laptop server computer. This computer is used to connect the on-board computer on the Boat. Power to this computer is provided from Black box, which is connected to a power supply.

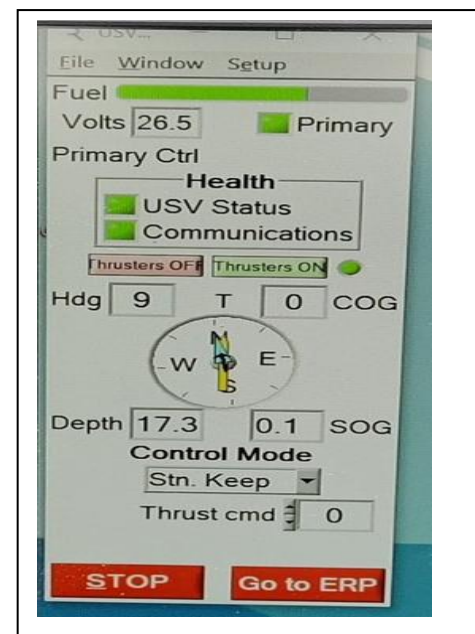
Login: **searobot**

Software tools installed on this computer include:

- **SeaRobot: SeaRobotics USV (Unmanned Surface Vehicle) v128**
<https://www.searobotics.com/products/autonomous-surface-vehicles>
- **TeamViewer**

Before starting SeaRobotics USV software, make sure that you have turned on the boat, and connected the controller with the computer by quickly pressing the mode button 3 of the controller twice and joystick button 1 once

SeaRobotics USV: This software is used to establish the connection of Boat and the server through the Radio device connection. Once all the connection is established and we can control the boat then all the buttons in the software would be Green. If there is any problem in Connection or Low Battery, then it will show either red or orange in the respective status and we won't be able to control the Boat.



TeamViewer: TeamViewer is used to control the onboard computer in Boat using the Server Computer. The software is installed in both the computers. We use the server computer staying in the shore to control the onboard computer. The ID and Password for controlling the on-board computer are:

ID: 192.168.0.68

Password: searobot

The Remote Controller is connected to this computer using USB port.

7. Computer onboard HyCAT and software

This computer is built on the boat HyCAT and is controlled by the user via TeamViewer. Two major software tools are installed on the onboard computer to navigate and control the HyCAT boat to perform survey tasks.

- **HyPACK**
- **River Surveyor Live**

HyPACK: This software is used to conduct onboard bathymetric and water quality data collection and post data processing. The survey border and survey lines can be defined in the HyPACK software, and measurements can be taken accordingly. This software is used for more automated mission planning.

River Surveyor Live (SonTek software): This software is used to conduct the river flow velocity, bathymetry and discharge surveys across a river transect. Navigation and measurement readings are taken by this software is manual and requires close attention of the operator. River surveyor live is free, and can be installed on any computer.

8. Remote Controller

A remote Controller with Joysticks is used for manually navigating and operating the HyCAT boat. The functions and modes of the remote controller buttons are described as below:

- The left joystick with Up and Down push makes the boat to move in forward or backward direction.
- The right joystick with Right and Left push makes the boat to turn right(clockwise) or turn left (anti-clockwise) direction.
- Button 9 and 11 are used for the speed control.

- Button 8 and 10 are used for the controlling direction (increasing or decreasing the angle)



There are several Control Modes, which are available.

- Thruster Mode (press down left analog, 1)
- Station Keep Red button(B/6)
- Hypack Mode (A/7)
- Heading Mode (Y/5)
- Standby Mode (X/4)

Thruster Mode: In thruster mode, we can control the boat manually. The joystick can be used for the controlling purpose. Thruster Mode is activated by single pressing the left Thumb button of Joystick or from drop down button of Control Mode in SeaRobot software. (1)

Station Keep Mode: In this mode, the boat keeps in stationary condition. This mode can be used when we want to keep our boat at a specific location to do point sampling. Even though in the flowing water, the boat will maintain its location with powered motor. The **red button** (B/6) on the Joystick triggers Station Keep mode. This mode can also be activated from the Control Mode in SeaRobot Software.

HyPack Mode: This mode is used to control the boat in the automatic mode via HyPACK software. While taking the survey measurements in HyPACK mode, the HyCAT boat will move along the survey lines pre-specified in HyPACK software with the pre-specified speed. You can start the HyCAT boat in HyPACK mode, after setting the survey lines you want it to collect the data. This mode is activated by the **green button** (A/7)

Heading Mode: This is a semi-automatic mode, in which the heading direction and speed are fixed. user. After specifying the direction and speed, the boat will move in the straight line according to the pre-determined direction and speed. This mode is activated by **yellow button** (Y/5). Can also be changed in SeaRobotics USV software.

Standby Mode: This mode is similar to the parking. The boat is made still and the boat moves along the speed of water. The boat motor is not operating in this case. This mode is similar to the

Station Keep Mode, but the motor is not in operation and the boat may move with the movement of water. This mode is activated by blue button (X/4)

9. Software packages

On laptop server computer:

- *SeaRobotics USV v-128*: for navigating and control the HyCAT boat
- *TeamViewer*: for accessing and control onboard computer

Computer onboard HyCAT

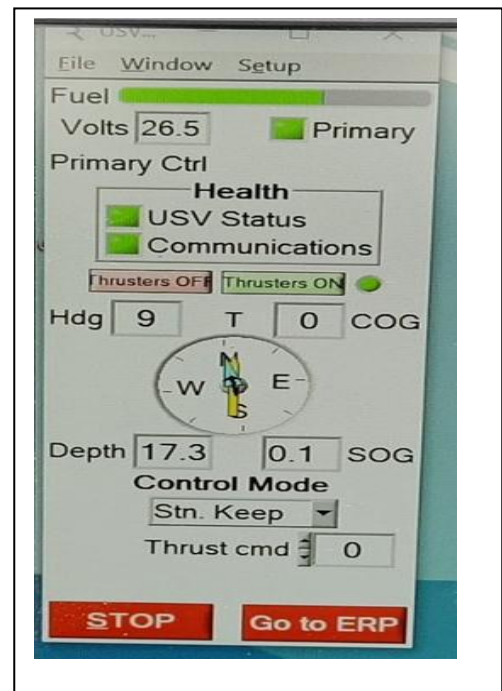
- HyPACK: for bathymetry and water quality survey with planned survey lines and border
- River Surveyor Live: river velocity and discharge survey

SeaRobotics USV software

It is installed on the laptop Server Computer. This is used to navigate the HYCAT boat and access HyCAT onboard computer to run River Surveyor Live and HyPACK software via TeamViewer connection. Once all the connection is established, all the buttons in the software main window would be Green and we can navigate and control the HyCAT boat. If there are any connection or Low Battery problems, the status buttons on the software main window will show either red or orange. In that case, we won't be able to navigate/control the HyCAT boat. This is the first software that we start, after we turned on the boat and established the connections.

The procedure for starting SeaRobotics software.

- Before starting this software, all the connections must be established between laptop computer, the blackbox hub, radio station, and the boat.
- Press the mode button (3) twice and click the left joystick button (1) once to turn on the controller and make in in thruster mode. Then, start the seaRobotics USV software on the Laptop computer.
- When all the status buttons on SeaRobotics software window are in green color, carry the HyCAT boat into the water for launching.
- SeaRobotics USV software shows the status about the Battery Health and connections.
- If there is any problem with the radio communication or the USV Health Status, the status buttons will show yellow or orange color. In that case, close SeaRobotics software and fix the problems. Then, restart the seaRobotics software. The above green status buttons signal that we are good to go.



HyPACK

HyPACK software is used to plan, collect and process Hydrographic and water quality data. This software is installed in the onboard computer. After the data acquisition of the survey is completed, remember to copy the data from onboard computer to the USB thumb drive on the HyCAT Boat and also copy it the laptop server computer. The collected raw data can be further processed in any computer with HYPACK software. Without the dongle key (Hypac key thumb drive) plugged in USB port of the computer, you can see the data, but cannot edit and process the data. The HyPACK has functions for mission planning, data collection, and post data processing. This software can ingest ESRI GIS shape files or KLM files to define the survey border or survey lines.

River Surveyor Live

This software is used for calculating the river discharge by measuring river flow velocity and the water depth across the riverbanks. It is installed on the onboard computer. The navigation of HyCAT boat with River Surveyor for river discharge measurements is manually controlled with the close attention of operator at start and end points. The collected data can be post-processed in River Surveyor Live and exported in the Matlab format or text format. Data can also be visualized and edited in River Surveyor.

Onboard Camera Control

The real-time video from the video camera onboard the HyCAT boat can be displayed on the laptop computer or TV monitor.

A bookmark of the Live camera footage can be found on the Google Chrome. If there are no bookmarks for the camera, then it can be opened from the following link:

192.168.0.20:8082/index.html

10. Collect River Discharge Data Using River Surveyor Live

Notes

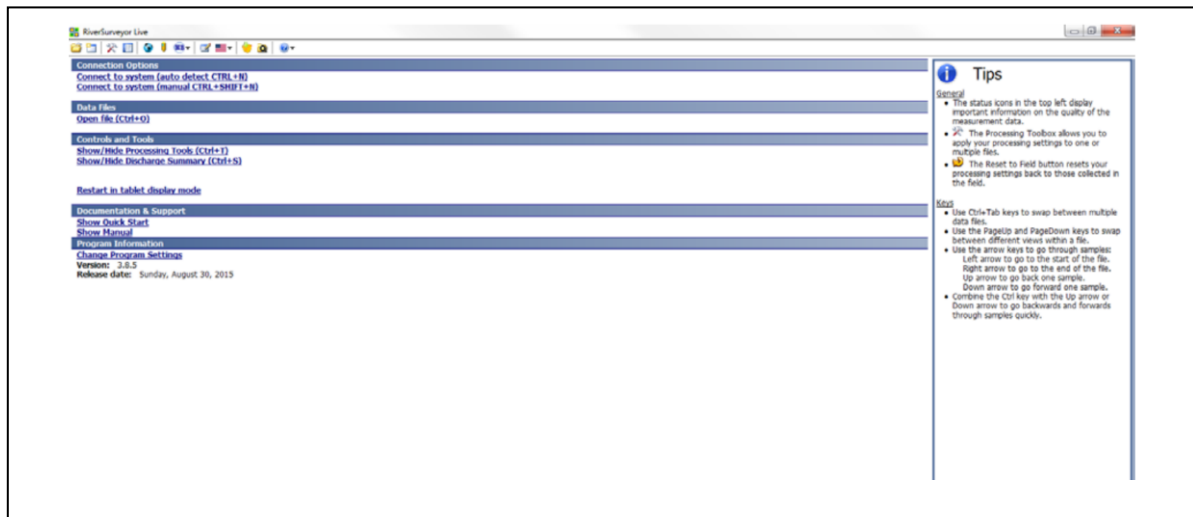
- Can output detailed velocity in Matlab format
- Velocity top and bottom have velocity gaps need to be extrapolated for discharge calculation
- Laser ranger is needed to define the distance from survey transect start point to the bank and the distance between the survey transect end point and the other bank
- Define the left and right bank by facing the flow direction
- Use River Surveyor Live to operate HyCAT manually, need to observe the HyCAT position to stop at the end point, and start the following line.
- Even survey passes (2, 4, 6), and 4 is recommended.
- River Surveyor training videos are available on Youtube at

<https://www.youtube.com/playlist?list=PL0537DDD4CAA81CA2>

Operation procedure

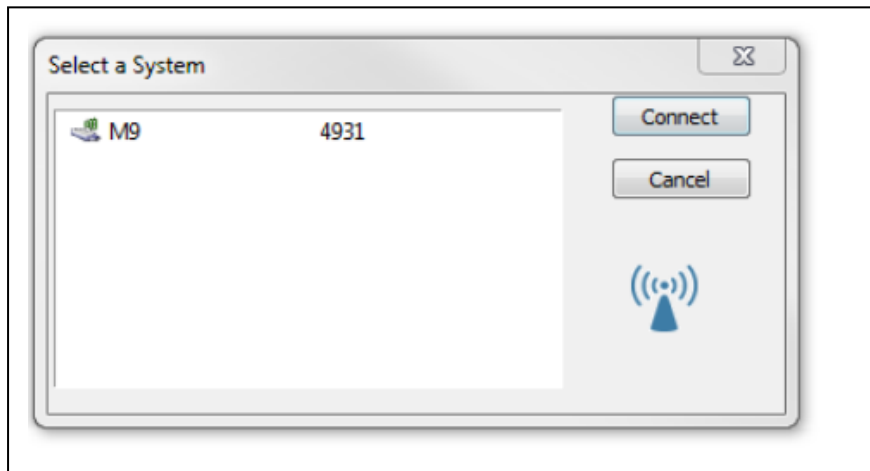
- Establish all the connections.
- Start the boat, start **seaRobotics** software and launch the boat.
- Manually navigate the boat with controller to the start point near the river.
- Do all the procedure mentioned for Manual Controlling of Boat.
- Open **River Surveyor Live**

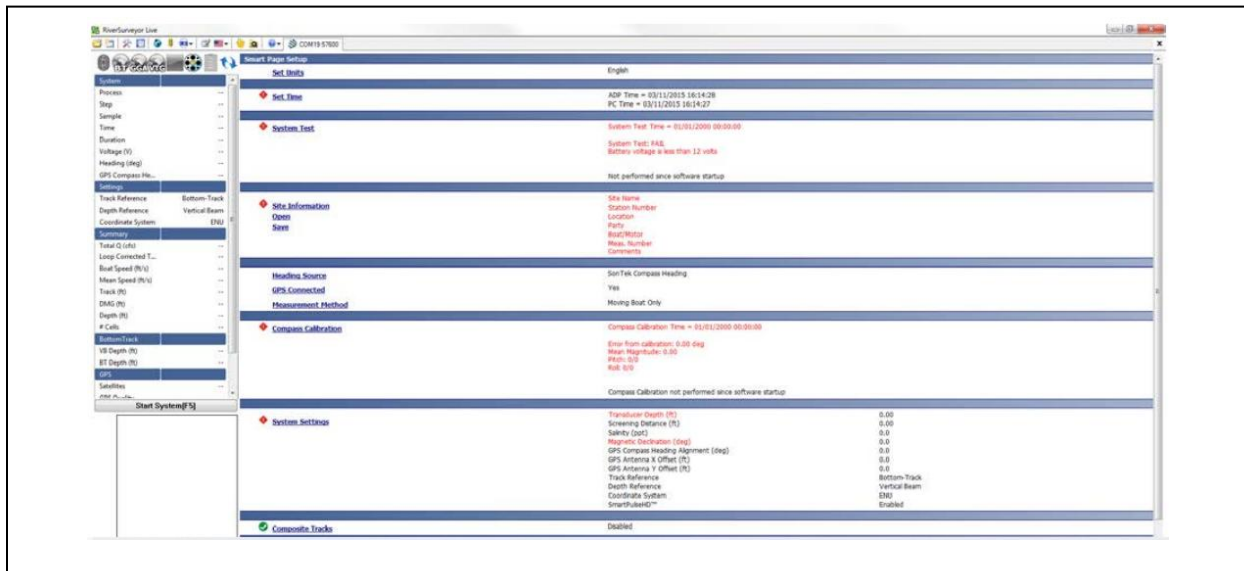
The main menu of River Surveyor is shown up:



Click on the **Connect** icon on the toolbar at the top of screen or use the automatic connection to establish the connection.

When the connection is established, the Smart Page is displayed as shown in figure below. This page allows to enter the Site Information, System Settings and Measurement Settings.





Site Information: Enter detailed information about the site.

System Settings:

Set the following parameters for the System Configuration.

Transducer Depth: 0.15

Screening Distance: 0.31

Magnetic Declination: -3.20 for Lake Tuscaloosa, which varies according to location and time

Track Reference: Bottom Track

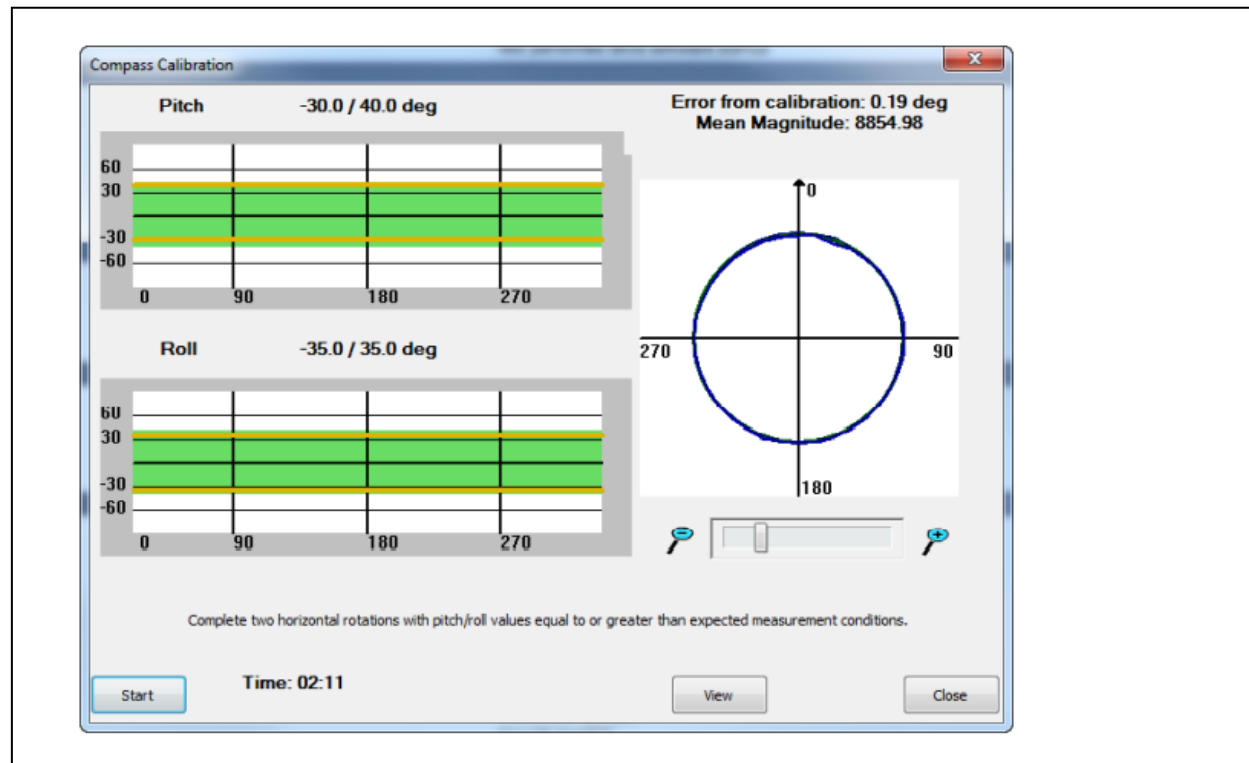
For other settings, keep default values

Compass Calibration:

To calibrate the boat compass, click on “Compass Calibration” on the Smart Page tab. The following window will appear. Click the Compass Calibration link and click Start. After two 360 rotations, the calibration is completed.

Compass Calibration window.

Click Stop upon completion of the compass calibration procedure. The calibration results would be displayed.



System Test:

The System Test performs a series of checks to confirm the battery, compass, recorder and temperature sensor are in good operating condition. Click Start to initiate the test. The test typically takes 60 seconds, and a dialog window will pop-up, indicating a system test Pass or Fail. A test failure would indicate that the user action is required to fix the problem and ensure proper functioning of the system. After the test is passed, we are good to go.

Set up Emergency Return Point, Transact Start and End Points

- Go to the Mission Status Page of the SeaRobotics software
- Under the River Gauge Setup Page, establish the Emergency return Point
- Navigate the boat to the location where you want to make it as the **Emergency Return Point** and click on USV button in the River Gauge Setup Page.
- After the Emergency Return Point is set up, we can now set the Starting and the Ending point of the River Transect.
 - Navigate the boat to the **start point** and click USV.
 - Drive the boat to the **end point** and click the USV button.
 - After the Emergency Return Point, Starting Point and Ending Point has been established, the information must to be sent to the boat.
 - Click on the **View Script** and then send the script to the boat.

- Now the Starting, Ending and Emergency Return Point is loaded to the on-board computer.

Take measurements along transect

- Click the Start a Measurement (F5) button to start data collection on the Smart Page of River Surveyor.
- Click **Start Edge** button and start collecting the samples. Input start edge dialog menu pops out, and press OK.
- When the boat reaches the end point, click **Stop Edge**. Start Edge again. Provide the Edge information.
- For this operation, the Mode is in **Waypoint mode**.

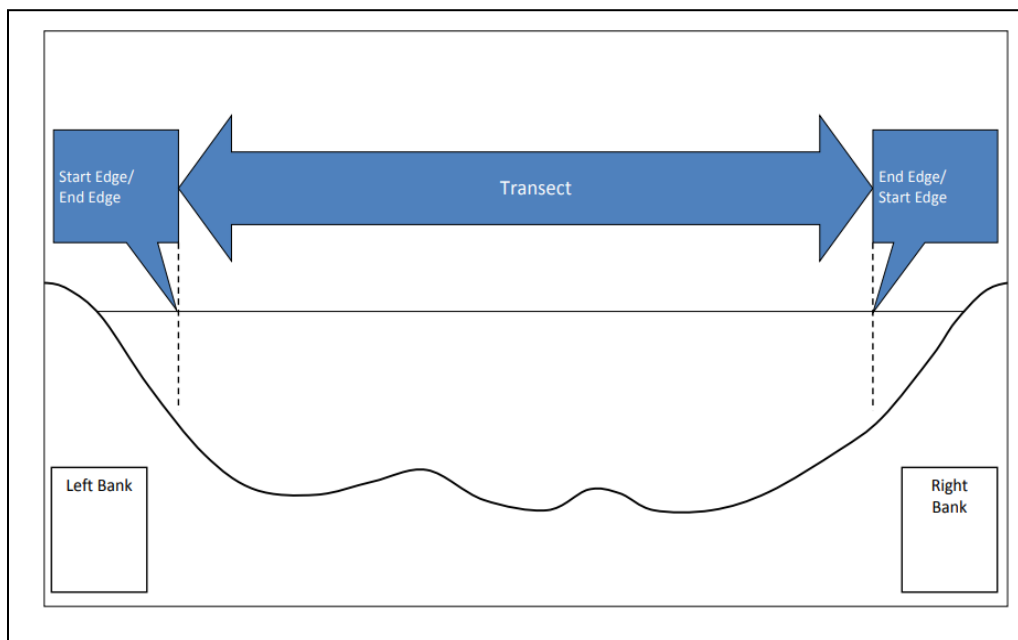
WP Station keep time: 45-60 second

Transects: 10 recommended

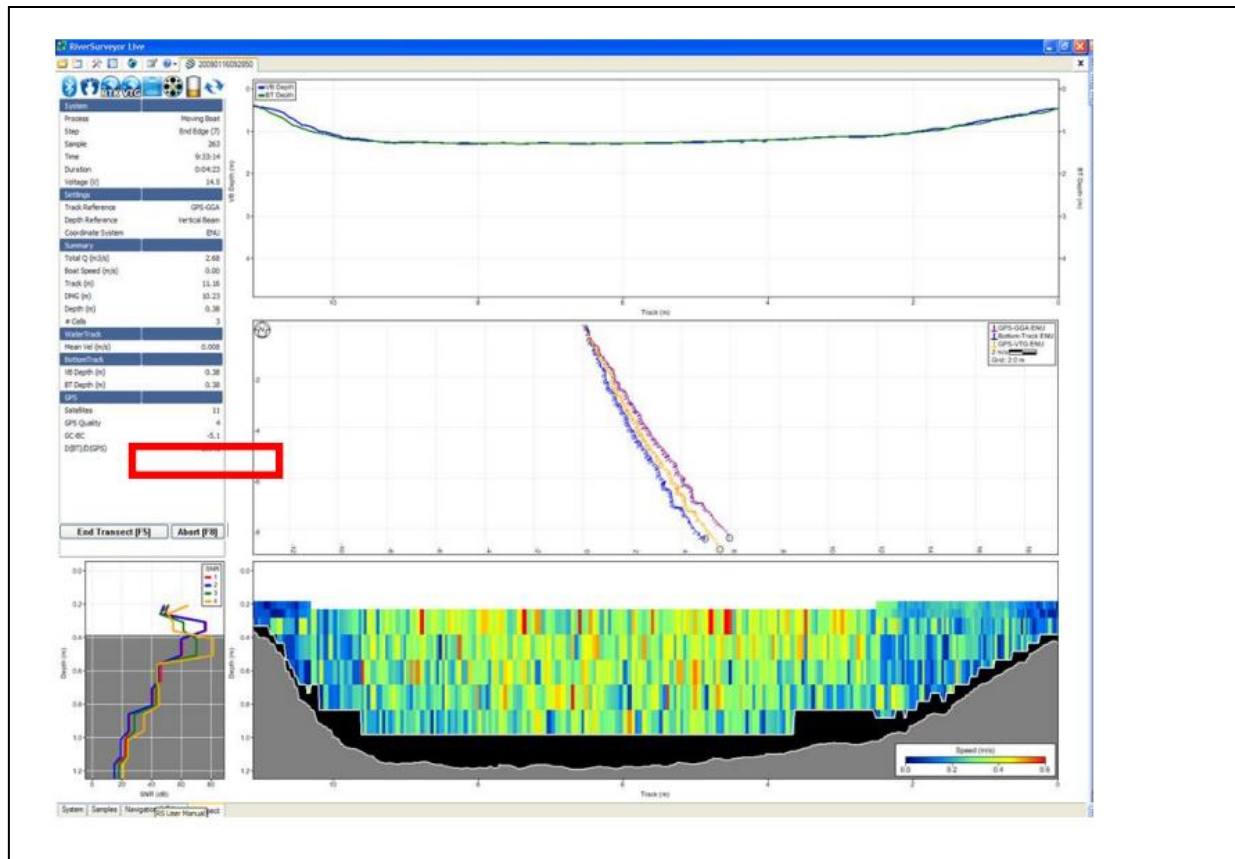
Thrust: According to need

The following diagram illustrate the measurement for calculation of velocity.

Transect:



Click the **End Transect** button upon the completion of the end edge. This automatically opens a new data collection window so you can start a new measurement. The system is still running, so if you need to make another measurement, click the Start Edge button (or F5) to start again. After the data collection is complete, click the **Abort** button or press F8 to stop. It is recommended to go to the **System** tab and download the recorded data files, upon completion of the survey



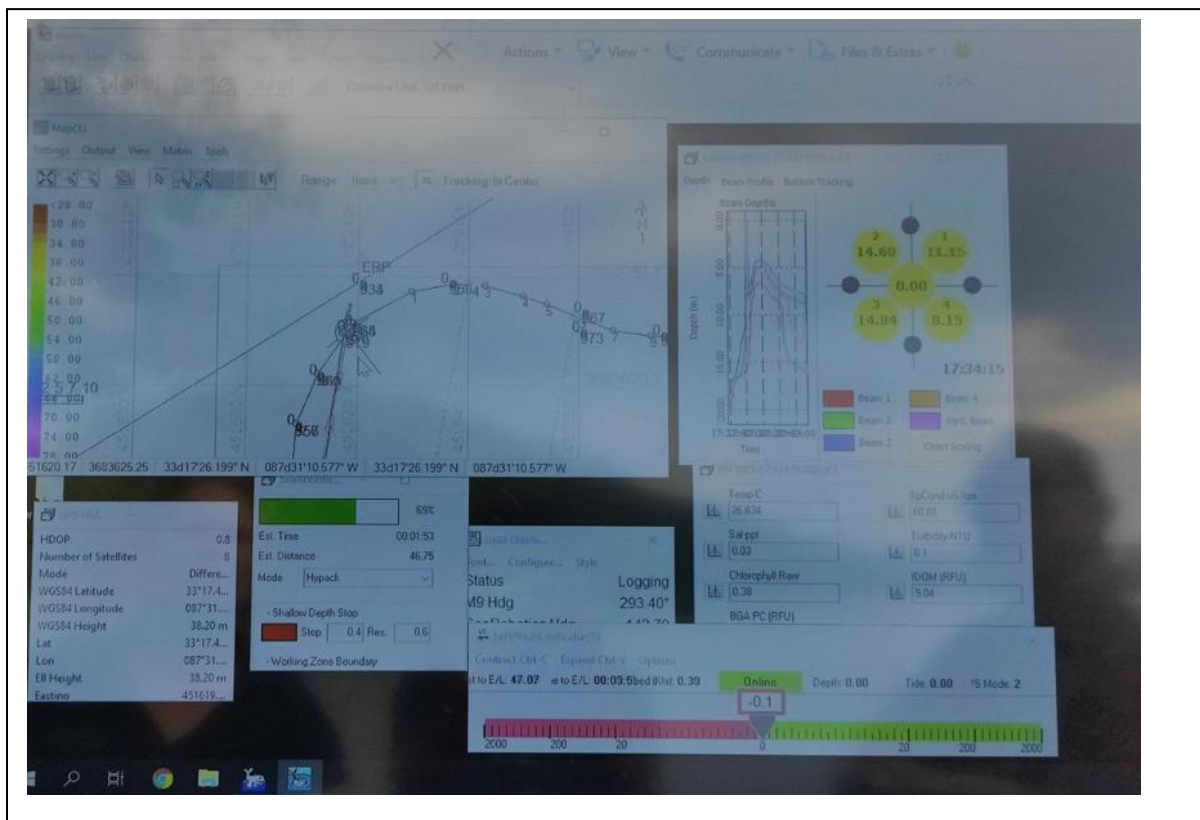
After all the transects are completed, we can display and look at the datasets in the River Surveyor itself.

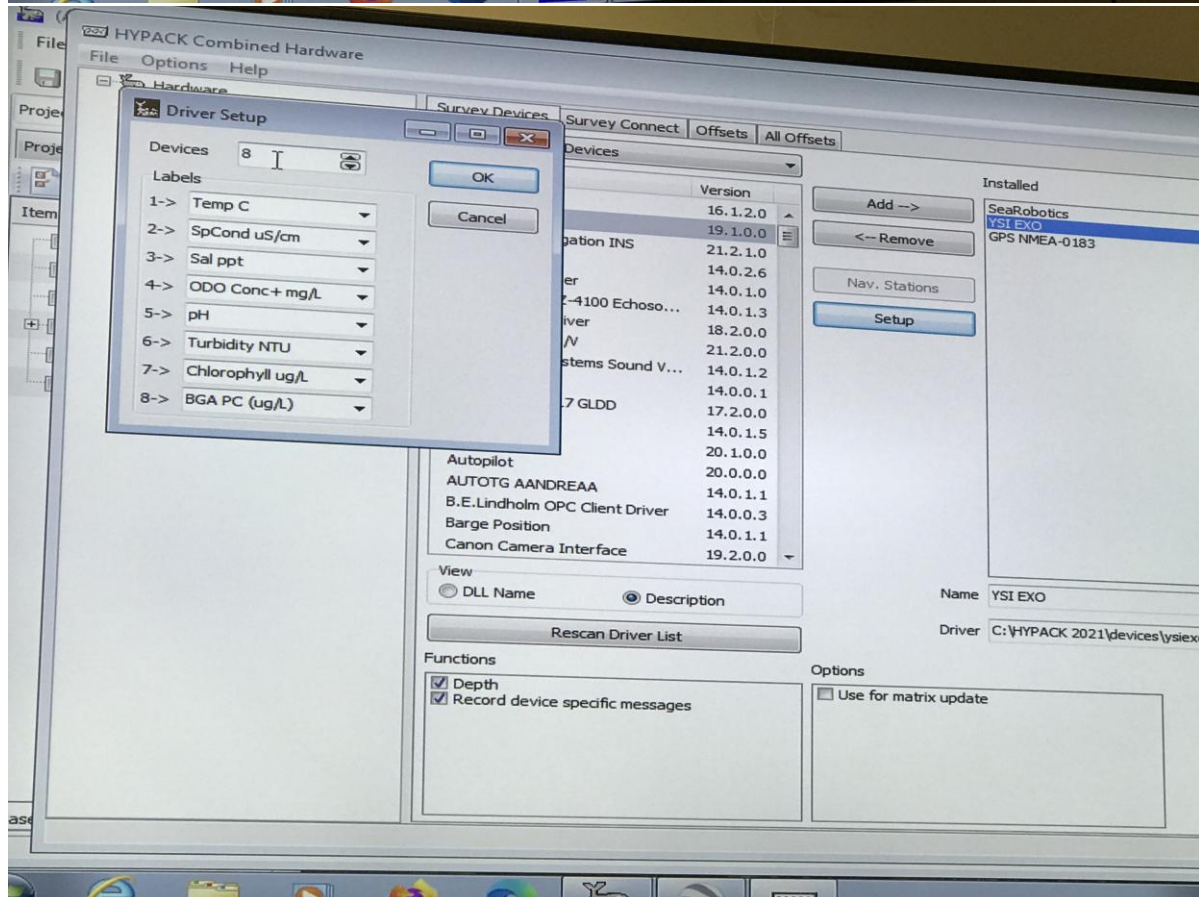
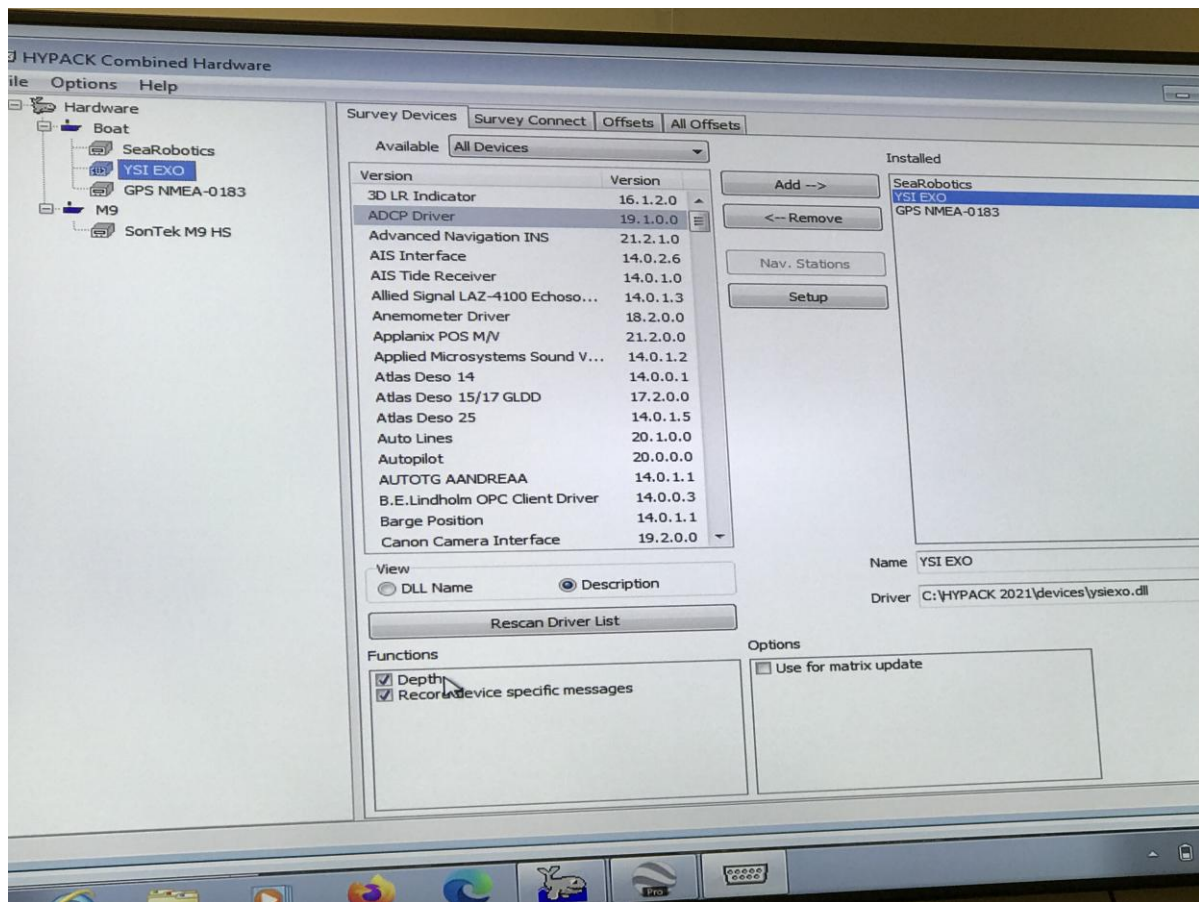
The information about the discharge can be observed from the Transect option. Also, take a look at Smart Page, System Samples Navigation Edges and Transect button.

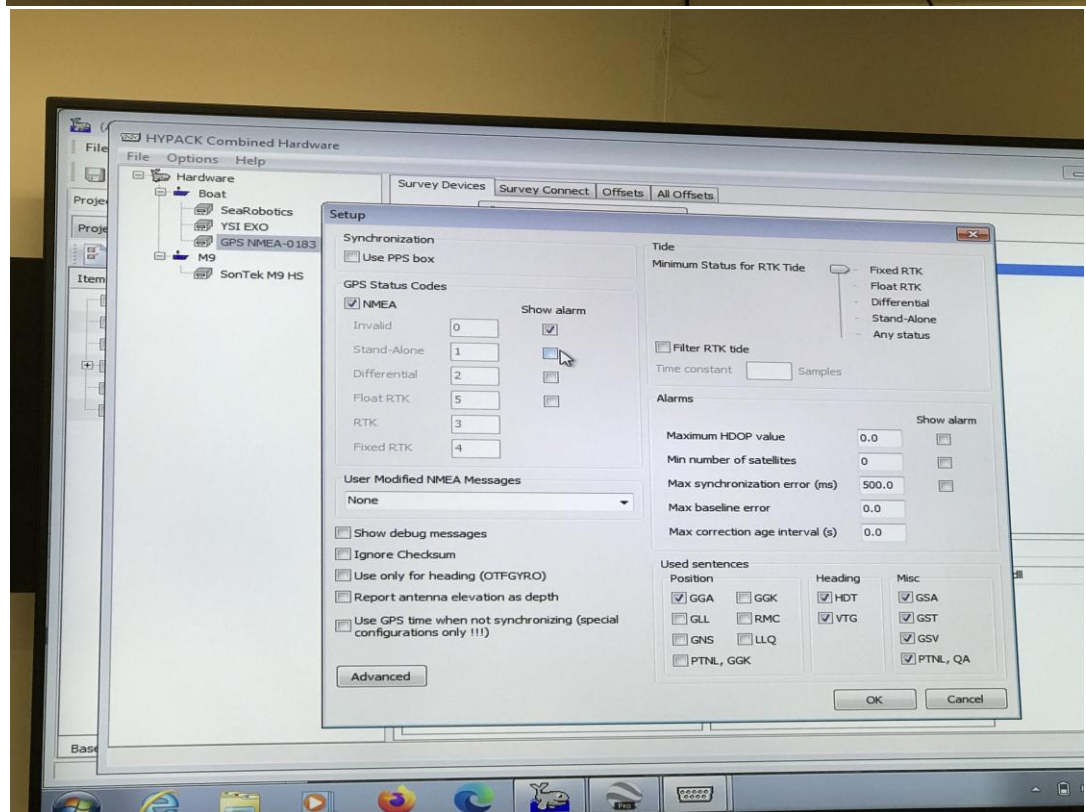
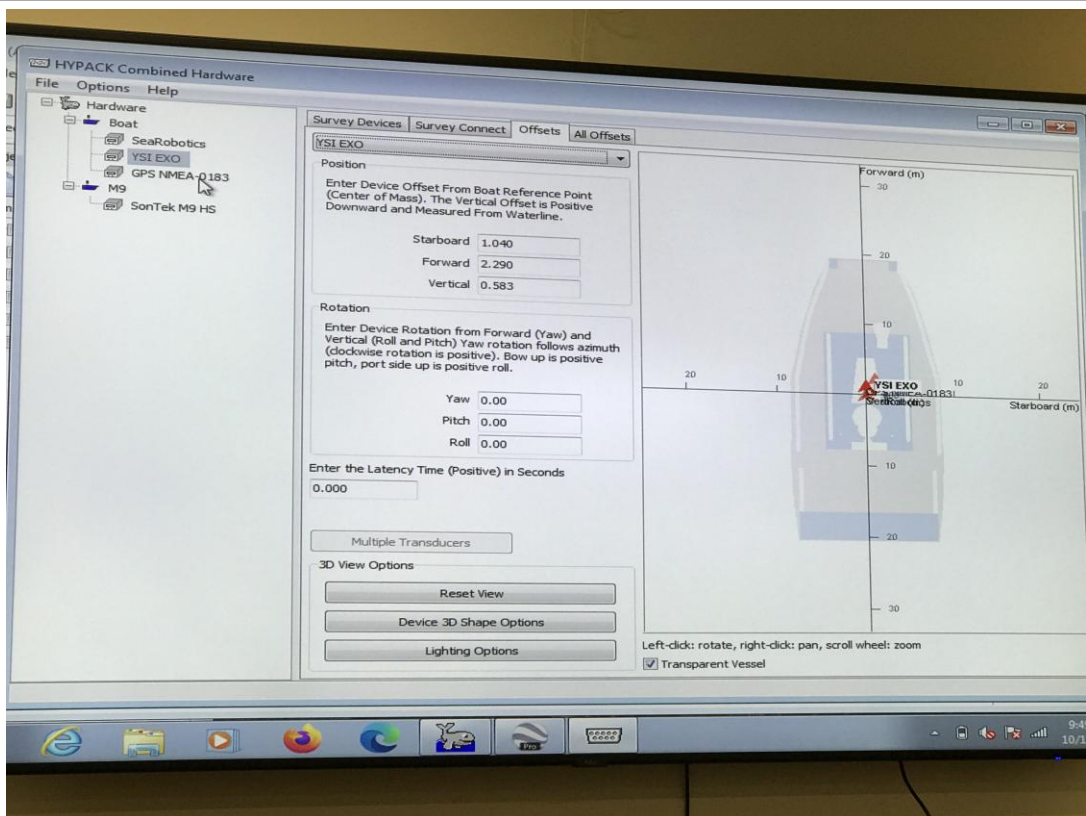
11. Collect Bathymetry and Water Quality Data using HyPACK

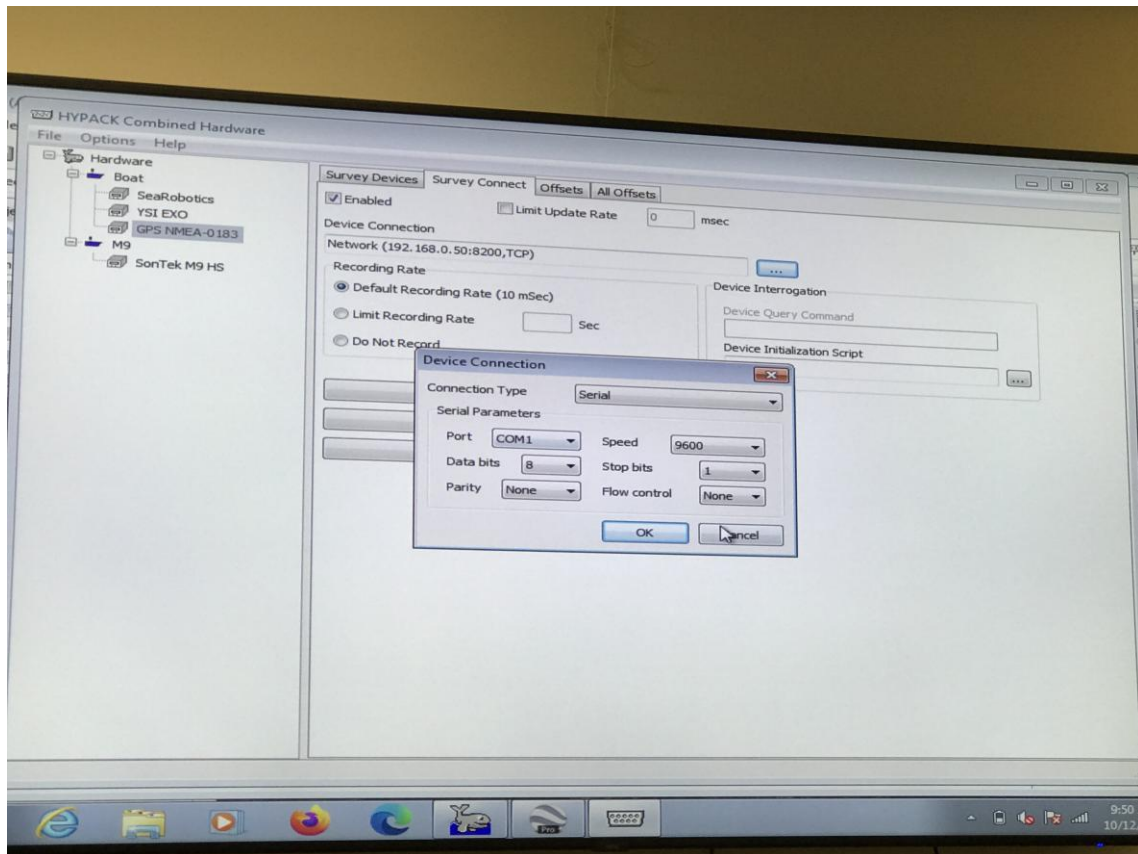
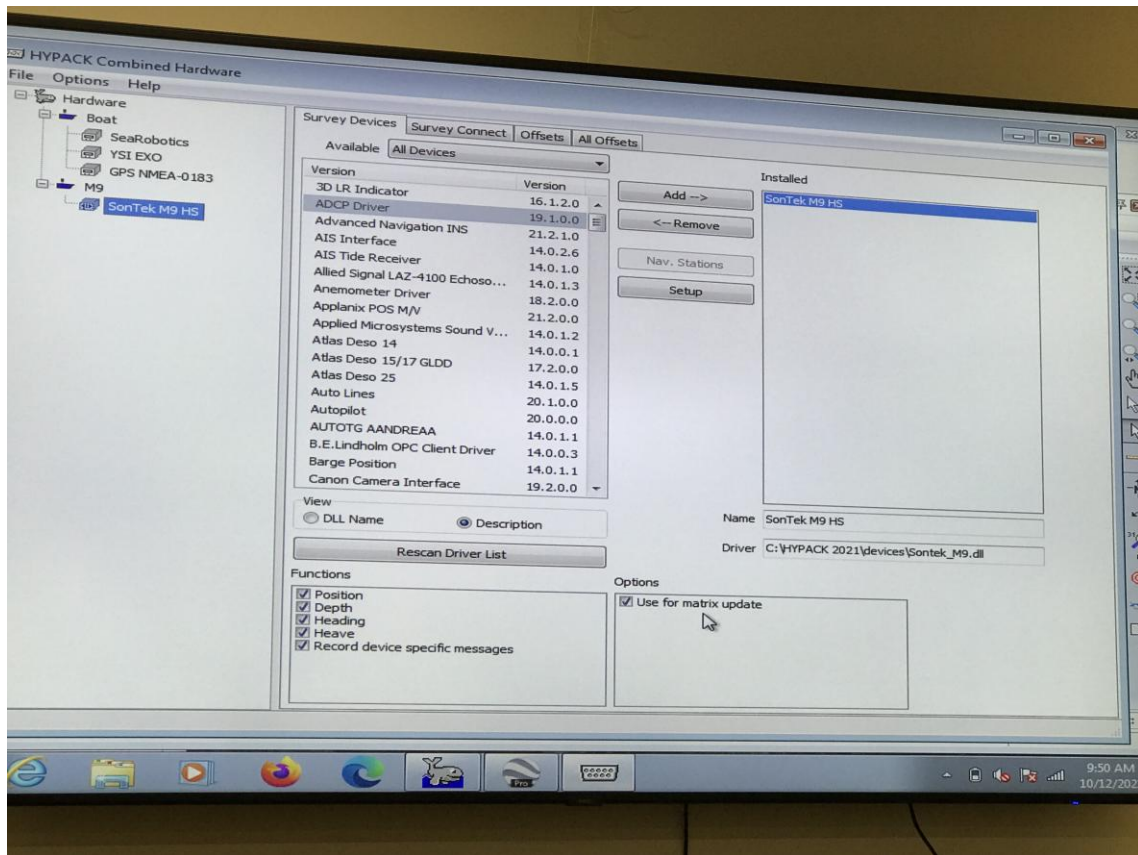
- Free training videos are available on Youtube at https://www.youtube.com/playlist?list=PLCbJJUmLdIBEK8_mEbmTQJSvWDOv9J12w
- The USB hardkey is required for boat operation and for HyPACK data processing
- HyPACK software is free and can be installed on any computer, but the data processing needs the hardkey plugged in.
- Start Boat, Radio, laptop Computer
- Make sure all the light in the Radio antenna receiver blink.

- Double Click the Mode and select the Thruster Mode
- Start the SeaRobotics software. Make sure all the status buttons are now in green.
- Establish Emergency Return Point.
- Open HyPACK 2020 Software
- Establish Geodesy
- Go to Settings Hardware: Check the Sea Robotics, YSI EXO, GPS NMEA-0183 and SonTek M9 HS, Define and connect each device: YSI EXO (com1), GPS NMEA-0183(2?), SonTek M9 HS(3?)
- M9 sampling rate 1 Hz, GPS sampling rate: 10 Hz, YSI to be set 1 Hz
- After we open a new project under the Folder, all the previous configuration parameters are uploaded to the new Project as the default values.
- Create and load a Google georeferenced image as background
- Use Border editor: On-screen digitizing or navigate HyCAT go along the shore, and use the trace to digitize to get border
- Use Line editor: draw the first line, then specify offset and line number and direction, or use the border to generate lines given the line azimuth direction and line spacing
- For the same survey lake site, the same project file can be repeatedly used with the planned border and lines, data raw files will be created with different file names (date and time), not overriding the previous files
- Load the Base Map, if desired
- Click Survey>Hypack or Click the Wheel button in the Toolbar.
- First, navigate the boat move along the boundary of the lake (the survey area), and create the border according to the boat trace/path
- Define the border and survey lines using Border Edit or Line Editor.
- Save the border file and line files to an appropriate file folder.
- A matrix can also be defined, depending on the need.
- After a survey line is selected, click **Start Survey**.
- A new Smart window appears, with a number of windows.
- After we make the survey lines and border ready, you must click **HYPACK** mode on the Remote Controller.
- It starts taking the measurements accordingly.
- You need to watch the Data Display Status menu, which should show LOGGING.
- Check all the reading, and make sure they are reasonable.









HyPack post processing

- Take look at the raw data using NotePad
- Raw data are in the separate text files, each file for one survey line, each raw file has the same setting header information
- Single beam correction
- Edit to delete outlier, etc
- After corrections and editing, save all line raw files as a single file
- Each survey line has five bathymetry beams, you can visualize all or each beam for editing
- If the editing is not complete, you can save as a special format for continued editing next time
- You can output xyz as a text file
- Water quality data, each parameter treat as a z to output xyz file separately.